

1. (2 points each) Each of the following multiple choice problems has one correct answer. Circle it. You do not need to show any reasoning.

(a) What is the radius of the sphere through $(-1, 2, 0)$ with center $(0, 0, 2)$?

(i) $\sqrt{2}$

(ii) 3

(iii) 4

(iv) $\sqrt{5}$

$$\text{radius} = \text{distance} = \sqrt{1^2 + 2^2 + 2^2} = 3$$

(b) If $\text{proj}_{\mathbf{a}} \mathbf{b} = \langle 1, 2, 3 \rangle$, what's $\text{proj}_{\langle 2, 4, 6 \rangle} \mathbf{b}$?

(i) $\langle 1, 2, 3 \rangle$

(ii) $\langle 2, 4, 6 \rangle$

(iii) $\langle 3, 6, 9 \rangle$

(iv) $\langle \sqrt{2}, 2\sqrt{2}, 3\sqrt{2} \rangle$

$\vec{a}$ and $\langle 1, 2, 3 \rangle$ are parallel, so these are the same!

(c) If $\mathbf{a}$ and $\mathbf{b}$ are unit vectors, which of the following cannot equal $\mathbf{a} \times \mathbf{b}$?

(i) $\langle 0, 0, 0 \rangle$

(ii) $\langle 0, 1, 0 \rangle$

(iii) $\langle 1, 0, 1 \rangle$

(iv) $\langle \frac{1}{2}, \frac{1}{2}, \frac{1}{2} \rangle$

$$|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta, \text{ so the magnitude must be } \leq 1.$$

$$\text{magnitude} = \sqrt{2}$$

(d) Which point lies on the line through $(1, 2, 3)$ and $(2, 0, 1)$?

(i) $(-1, 5, 7)$

(ii) $(0, 4, 5)$

(iii) $(2, 1, 2)$

(iv) $(3, -2, 5)$

$(1, 2, 3)$ is half way between this and $(2, 0, 1)$

(e) Which of the following planes is parallel to the line $\mathbf{r}(t) = \langle t, 2t, 3t \rangle$?

(i) $x + y + z = 6$

(ii) $x + 2y + 3z = 1$

(iii) $2x - y + z = 4$

(iv) $x + y - z = 7$

Normal vector should be orthogonal to $\langle 1, 2, 3 \rangle$

$$\langle 1, 2, 3 \rangle \cdot \langle 1, 1, -1 \rangle = 0$$

2. (6 points per part) **Parts (a) and (b) are unrelated.**

- (a) Find an equation for the surface consisting of all points (x, y, z) such that the distance from (x, y, z) to the x -axis is equal to the distance from (x, y, z) to the point $(0, 0, 2)$, and give a precise name for the corresponding 3D surface.

$$\sqrt{y^2 + z^2} = \sqrt{x^2 + y^2 + (z-2)^2}$$

$$\cancel{y^2} + \cancel{z^2} = x^2 + \cancel{y^2} + \cancel{z^2} - 4z + 4$$

$$4z = x^2 + 4$$

Equation:

$$4z = x^2 + 4$$

Name:

Parabolic cylinder

- (b) Find a vector function $\mathbf{r}(t)$ whose space curve is the curve of intersection of the cylinder $y^2 + z^2 = 1$ and the hyperbolic paraboloid $x - y^2 + z^2 = 2$.

$$y = \cos t$$

$$z = \sin t$$

$$x = 2 + y^2 - z^2$$

$$x = 2 + \cos^2 t - \sin^2 t$$

$\mathbf{r}(t) =$

$$\langle 2 + \cos^2 t - \sin^2 t, \cos t, \sin t \rangle$$

3. (6 points per part) For parts (a) and (b), let $\mathcal{S}$ be the surface $z = \frac{y}{x} + \sqrt{y-2x}$.

(a) Find the equation of the plane tangent to $\mathcal{S}$ at $(2, 8, 6)$.

$$\left. \begin{aligned} \frac{\partial z}{\partial x} &= \frac{-y}{x^2} - \frac{1}{\sqrt{y-2x}} \\ \frac{\partial z}{\partial y} &= \frac{1}{x} + \frac{1}{2\sqrt{y-2x}} \end{aligned} \right\} \begin{matrix} x=2 \\ y=8 \end{matrix} \quad \begin{aligned} \frac{\partial z}{\partial x} &= -\frac{5}{2} \\ \frac{\partial z}{\partial y} &= \frac{3}{4} \end{aligned}$$

Plane equation:

$$z = -\frac{5}{2}(x-2) + \frac{3}{4}(y-8) + 6$$

(b) Use linearization to estimate the x -coordinate of the point $(x, 7.98, 5.91)$ on $\mathcal{S}$.

$$5.91 = -\frac{5}{2}(x-2) + \frac{3}{4}(7.98-8) + 6$$

$$-\frac{5}{2}(x-2) = -0.075$$

$$x \approx \boxed{x = 2.03}$$

4. (13 points) Find the volume of the largest rectangular box in the first octant with three faces in the coordinate planes and the vertex opposite the origin (i.e. the one marked P in the picture below) on the surface $2x^2 + y + z = 90$.

In order to receive full credit, you must show that your answer really is the maximum.

Let $P = (x, y, z)$, so volume is

$$V = xyz \text{ with } 2x^2 + y + z = 90$$

$$z = 90 - 2x^2 - y$$

Want max of

$$f(x, y) = xy(90 - 2x^2 - y)$$

$$= 90xy - 2x^3y - xy^2$$

$$f_x(x, y) = 90y - 6x^2y - y^2 = 0$$

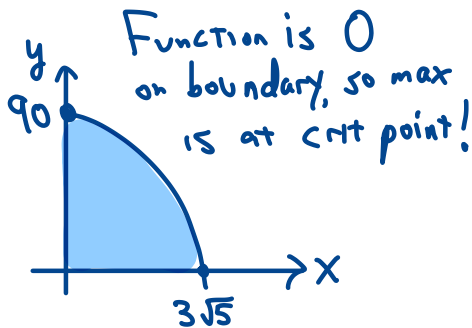
$$f_y(x, y) = 90x - 2x^3 - 2xy = 0$$

ok to divide by x or y ,
b/c x or y will never be
zero @ a max volume

$$90 - 6x^2 - y = 0 \rightarrow y = 90 - 6x^2$$

$$90 - 2x^2 - 2y = 0 \rightarrow 90 - 2x^2 - 2(90 - 6x^2) = 0$$

This is the max
because the domain
 $x \geq 0, y \geq 0, 90 - 2x^2 - y \geq 0$
is closed & bounded.



$$10x^2 = 90$$

$$x^2 = 9$$

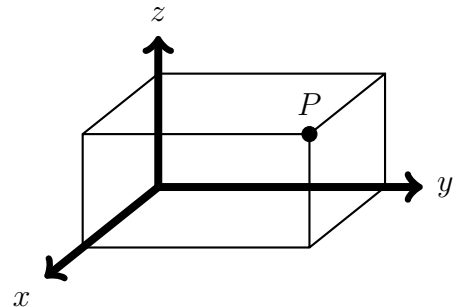
$$x = 3$$

positive b/c first
OCTANT

$$y = 90 - 6(3)^2 = 36$$

$$z = 90 - 2(3)^2 - 36 = 36$$

$$\text{So } x = 3, y = 36, z = 36$$



Maximum volume:

3888

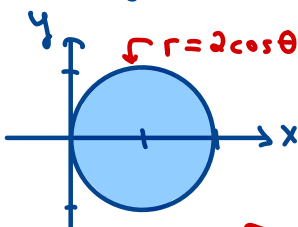
5. (12 points) Set up, but **do not evaluate**, an iterated integral using polar coordinates equal to the volume between the paraboloids $z = -2x + 3x^2 + 3y^2$ and $z = 2x^2 + 2y^2$.

Intersection is
 $-2x + 3x^2 + 3y^2 = 2x^2 + 2y^2$
 $x^2 - 2x + 1 + y^2 = 0 + 1$
 $(x-1)^2 + y^2 = 1 \rightarrow$

In this region, $z = 2x^2 + 2y^2$
 is above $z = -2x + 3x^2 + 3y^2$.
 (can test at a point like (1,0))

So integrand is $(2x^2 + 2y^2) - (-2x + 3x^2 + 3y^2)$
 $\downarrow$ polar
 $2r^2 - (-2r\cos\theta + 3r^2) = 2r\cos\theta - r^2$

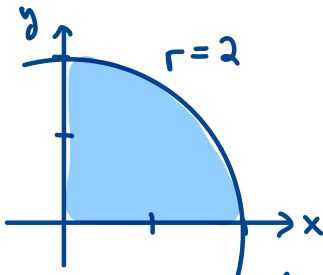
Region of integration:
 $r = 2\cos\theta$
 $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$
 $0 \leq r \leq 2\cos\theta$



Integral:

$$\int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} (2r\cos\theta - r^2) r \, dr \, d\theta$$

6. (13 points) Find the center of mass of the lamina that occupies the quarter disk in the first quadrant bounded by $x^2 + y^2 = 4$ with density function $\rho(x, y) = 1 + x^2 + y^2$.



Note: region and
density function are
symmetric in x & y ,
so $\bar{x} = \bar{y}$.

$$\begin{aligned} \text{Mass: } m &= \int_0^{\pi/2} \int_0^2 (1+r^2) r dr d\theta \\ &= \int_0^{\pi/2} \left(\frac{1}{2} r^2 + \frac{1}{4} r^4 \right) \Big|_{r=0}^{r=2} d\theta = \int_0^{\pi/2} (2+4) d\theta = 6\theta \Big|_0^{\pi/2} = 3\pi \end{aligned}$$

$$\begin{aligned} M_y &= \int_0^{\pi/2} \int_0^2 (1+r^2) (\overbrace{r \cos \theta}^x) r dr d\theta = \int_0^{\pi/2} \cos \theta \left(\int_0^2 (r^2 + r^4) dr \right) d\theta \\ &= \int_0^{\pi/2} \cos \theta \left(\frac{1}{3} r^3 + \frac{1}{5} r^5 \right) \Big|_{r=0}^{r=2} d\theta = \int_0^{\pi/2} \cos \theta \left(\frac{8}{3} + \frac{32}{5} \right) d\theta \\ &= \frac{136}{15} \int_0^{\pi/2} \cos \theta d\theta = \frac{136}{15} (\sin \theta) \Big|_0^{\pi/2} = \frac{136}{15} \end{aligned}$$

$$x = \frac{\frac{136}{15}}{3\pi} = \frac{136}{45\pi}, \text{ and } \bar{y} \text{ is the same!}$$

$$(\bar{x}, \bar{y}) = \left(\frac{136}{45\pi}, \frac{136}{45\pi} \right)$$

7. (14 points) Let $g(x) = \frac{1}{2} \ln(2x - 3)$.

In this problem you will find the Taylor polynomial based at $b = 2$ (**NOT** at $b = 0$).

(a) Find the third Taylor polynomial, $T_3(x)$, for the function g based at $b = 2$.

$$\begin{aligned} g(x) &= \frac{1}{2} \ln(2x-3) & g(2) &= 0 \\ g'(x) &= \frac{1}{2x-3} & g'(2) &= 1 \\ g''(x) &= \frac{-2}{(2x-3)^2} & g''(2) &= -2 \\ g'''(x) &= \frac{8}{(2x-3)^3} & g'''(2) &= 8 \end{aligned}$$

$$T_3(x) = (x-2) - (x-2)^2 + \frac{4}{3}(x-2)^3$$

(b) Use Taylor's inequality to find an upper bound (as sharp as possible) for $|f(x) - T_3(x)|$ on the interval $[1.8, 2.2]$.

$$\left| g^{(4)}(x) \right| = \left| \frac{-48}{(2x-3)^4} \right| \quad \text{largest when denominator is smallest, at } x = 1.8$$

$$M = \frac{48}{(0.6)^4} = \frac{10000}{27}$$

$$\left| T_3(x) - f(x) \right| \leq \frac{1}{24} \cdot \frac{10000}{27} (0.2)^4 = \frac{2}{81}$$

$$\approx 0.02469$$

$$\frac{2}{81}$$

Upper bound: _____

8. (14 points) Let $f(x) = x \sin(2x^3)$.

(a) Find the Taylor series for $f(x)$ based at 0.

Simplify your final answer and write it in sigma notation.

$$\begin{aligned} \sin(x) &= \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} \\ &\quad \downarrow \text{replace } x \text{ w/ } 2x^3 \\ \sin(2x^3) &= \sum_{k=0}^{\infty} \frac{(-1)^k 2^{2k+1} x^{6k+3}}{(2k+1)!} \\ &\quad \downarrow \text{mult. by } x \\ x \sin(2x^3) &= \sum_{k=0}^{\infty} \frac{(-1)^k 2^{2k+1} x^{6k+4}}{(2k+1)!} \end{aligned}$$

Taylor series: _____

$$\sum_{k=0}^{\infty} \frac{(-1)^k 2^{2k+1} x^{6k+4}}{(2k+1)!}$$

(b) Find $f^{(16)}(0)$.

Want x^{16} term of T. series. $6k+4=16 \rightarrow k=2$

This term is $\frac{2^5 x^{16}}{5!} = \frac{f^{(16)}(0) x^{16}}{16!}$ (by defn of Taylor series)

$$\frac{2^5}{5!} = \frac{f^{(16)}(0)}{16!}$$

$f^{(16)}(0) =$ _____

$$\frac{32 \cdot 16!}{5!}$$

(c) Give the first two nonzero terms of the Taylor series for $g(t) = \int_0^t x \sin(2x^3) dx$.

$$\begin{aligned} g(t) &\approx \int_0^t \left(\frac{2x^4}{1} - \frac{8x^{10}}{6} \right) dx \\ &= \left(\frac{2}{5} x^5 - \frac{4}{33} x^{11} \right) \Big|_0^t \end{aligned}$$

$$\frac{2}{5} t^5 - \frac{4}{33} t^{11}$$

First two nonzero terms: _____